

Cell lines immunopeptidome analysis focused on CTAs associated with overall survival - Figure 4 - Supplementary file 2

Léa ROGUE

2026-02-16

List of Figures

1	Venn plot JJ012 (MHCI)	7
2	Venn plot CH2879 (MHCI)	8
3	Heatmap of abund	11
4	Scatter plot from Fig1E (CH2879)	14
5	Rank scatter plot (CH2879)	16
6	Scatter plot from Fig1E (JJ012)	19
7	Rank scatter plot (JJ012)	21

Contents

Load packages	4
Load functions	4
Read data	5
I. Venn diagram with CTAs	6
JJ	7
CH	8
II. RNA and protein abundances in cell lines	9
RNA	9
Protein abundances	9
Heatmap	10
III. Peptide ranks and scatter plots	12
CH2879	12
Scatter plot from Fig1E	12
Rank scatter plot	15
JJ012	17
Scatter plot from Fig1E	17
Rank scatter plot	20
IV. Supplementary data 2	22

Load packages

```
library(ComplexHeatmap)
library(circlize)
library(grid)
library(ggplot2)
library(reshape2)
library(ggvenn)
library(UniProt.ws)
library(dplyr)
library(tidyr)
library(VennDiagram)
library(plotly)
library(ggrepel)
library(writexl)
```

Load functions

```
map_genes_to_peptides <- function(df_summary) {
  df_sep <- df_summary %>%
    mutate(RowID = row_number()) %>%
    separate_rows(Prot_accessions, sep = ";\s*") %>%
    mutate(Prot_accessions = sub("-\\d+$", "", Prot_accessions))

  gene_mapping <- AnnotationDbi::select(up, keys = unique(df_sep$Prot_accessions),
    columns = "gene_primary", keytype = "UniProtKB")

  df_merged <- df_sep %>%
    left_join(gene_mapping, by = c(Prot_accessions = "Entry"))

  df_final <- df_merged %>%
    group_by(Seq) %>%
    summarise(Proteins = paste(unique(Prot_accessions), collapse = "; "),
      Genes = paste(unique(Gene.Names..primary.), collapse = "; "),
      .groups = "drop")

  df_summary %>%
    left_join(df_final, by = "Seq") %>%
    filter(Genes %in% l_CTA)
}

prep_df <- function(df) {
  df_summary <- df %>%
    group_by(Prot_accessions) %>%
    summarise(nb_sequences = n(), .groups = "drop")

  # Separate
  df_sep <- df_summary %>%
    mutate(RowID = row_number()) %>%
    separate_rows(Prot_accessions, sep = ";\s*") %>%

```

```

mutate(Prot_accessions = sub("-\\d+$", "", Prot_accessions))

# Mapping UniProt -> Gene(s)
gene_mapping <- AnnotationDbi::select(up, keys = df_sep$Prot_accessions,
  columns = "gene_primary", keytype = "UniProtKB")

# Merge and reconstruct
df_final <- df_sep %>%
  left_join(gene_mapping, by = c(Prot_accessions = "Entry")) %>%
  group_by(RowID, nb_sequences) %>%
  summarise(Proteins = paste(unique(Prot_accessions), collapse = "; "),
    Genes = paste(unique(Gene.Names$.primary.), collapse = "; "),
    .groups = "drop") %>%
  arrange(desc(nb_sequences))

df_final <- df_final %>%
  group_by(Genes) %>%
  mutate(Genes_unique = if (n() > 1)
    paste0(Genes, "_", row_number()) else Genes) %>%
  ungroup() %>%
  mutate(Genes_unique = factor(Genes_unique, levels = Genes_unique)) %>%
  arrange(desc(nb_sequences))
return(df_final)
}

```

Read data

```

# Cell lines immunopeptidomic
df_ch_mhc1 <- read.table("../data/immunoep_CH2879.tsv", sep = "\t",
  header = T)
df_jj_mhc1 <- read.table("../data/immunoep_JJ012.tsv", sep = "\t",
  header = T)

# Filter on length
df_jj_mhc1 <- df_jj_mhc1[df_jj_mhc1$seq_len >= 8 & df_jj_mhc1$seq_len <=
  11, ]
df_ch_mhc1 <- df_ch_mhc1[df_ch_mhc1$seq_len >= 8 & df_ch_mhc1$seq_len <=
  11, ]

# CTA
l_CTA <- read.table("../results/clusters_cta_signif_conv_indiv.tsv",
  header = T)$SYMBOL
l_clust <- read.table("../results/clusters_cta_signif_conv_indiv.tsv",
  sep = "\t", header = T)

# Read IEDB file
df_iedb_mhc1 <- read.csv("../data/epitope_table_export_human_MHC_I.tsv",
  header = T, sep = "\t")
df_pep_mhc1 <- df_iedb_mhc1[, c("Name", "Source.Molecule.IRI",
  "Molecule.Parent.IRI")]

```

```

# IEDB
pep_iedb_mhc1 <- unique(df_pep_mhc1$Name)
pep_iedb_mhc1 <- gsub(".*|\\++.*", "", pep_iedb_mhc1)

# Read scatter data from Fig1E
df_scatter_hr <- read.table("../results/table_scatter_63_complete.tsv",
  sep = "\t", header = T)

# Read and prepare strong binders files
df_aff_jj <- read.table("../data/table_res_netmhc_all_JJ012.tsv",
  sep = "\t", header = T)
df_aff_ch <- read.table("../data/table_res_netmhc_all_CH2879.tsv",
  sep = "\t", header = T)

# Read TPM
df_ch2879_tpm <- read.table("../results/CH2879_fpk_m_tpm.tsv",
  sep = "\t", header = T)
df_jj012_tpm <- read.table("../results/JJ012_fpk_m_tpm.tsv", sep = "\t",
  header = T)

# Read prot abundances
df_prot <- read.table("../data/immuno-peptide-proteo.tsv", sep = "\t",
  header = T)

```

I. Venn diagram with CTAs

```

# CTA iedb
df_iedb_mhc1$ID <- gsub("http://www.uniprot.org/uniprot/", "",
  df_iedb_mhc1$Molecule.Parent.IRI)
df_iedb_mhc1$ID <- sub("-.*", "", df_iedb_mhc1$ID)

# Transform ID in gene name
up <- UniProt.ws(taxId = 9606)
gene_mapping <- AnnotationDbi::select(up, keys = unique(df_iedb_mhc1$ID[df_iedb_mhc1$ID !=
  ""]), columns = "gene_primary", keytype = "UniProtKB")
colnames(gene_mapping) <- c("From", "ID", "Gene.Names..primary.")
df_iedb_mhc1_name <- merge(df_iedb_mhc1, gene_mapping, by.x = "ID",
  all.x = TRUE)

# Select peptides from CTAs of IEDB
df_iedb_mhc1_cta <- df_iedb_mhc1_name[df_iedb_mhc1_name$Gene.Names..primary. %in%
  1_CTA, ]

# Take CTAs from peptides in cell lines
df_final_pep_jj <- map_genes_to_peptides(df_jj_mhc1)
df_final_pep_ch <- map_genes_to_peptides(df_ch_mhc1)

# Clean pep iedb
pep_iedb_mhc1_cta <- unique(df_iedb_mhc1_cta$Name)
pep_iedb_mhc1_cta <- gsub(".*|\\++.*", "", pep_iedb_mhc1_cta)

```

```

# Select new pep CH
new_pep_ch_mhc1 <- setdiff(df_final_pep_ch$Seq, pep_iedb_mhc1_cta)
grep_ch <- sapply(new_pep_ch_mhc1, function(x) any(grepl(x, pep_iedb_mhc1_cta)))
new_pep_ch_mhc1 <- new_pep_ch_mhc1[!new_pep_ch_mhc1 %in% names(grep_ch)[grep_ch]]
df_ch_new_pep_mhc1 <- df_ch_mhc1 %>%
  filter(Seq %in% new_pep_ch_mhc1) %>%
  distinct(Seq, .keep_all = TRUE)

# Select new pep JJ
new_pep_jj_mhc1 <- setdiff(df_final_pep_jj$Seq, pep_iedb_mhc1_cta)
grep_jj <- sapply(new_pep_jj_mhc1, function(x) any(grepl(x, pep_iedb_mhc1_cta)))
new_pep_jj_mhc1 <- new_pep_jj_mhc1[!new_pep_jj_mhc1 %in% names(grep_jj)[grep_jj]]

df_jj_new_pep_mhc1 <- df_jj_mhc1 %>%
  filter(Seq %in% new_pep_jj_mhc1) %>%
  distinct(Seq, .keep_all = TRUE)

# Clean IEDB pep
pep_iedb_extended_ch <- unique(c(pep_iedb_mhc1_cta, names(grep_ch)[grep_ch]))
pep_iedb_extended_jj <- unique(c(pep_iedb_mhc1_cta, names(grep_jj)[grep_jj]))

```

JJ

```

# Venn diagram
grid.newpage()
venn <- draw.pairwise.venn(area1 = length(pep_iedb_extended_jj),
  area2 = length(df_final_pep_jj$Seq), cross.area = length(df_final_pep_jj$Seq) -
    length(new_pep_jj_mhc1), category = c("IEDB", "JJ012"),
  fill = c("#2C7BB6", "#ff7f0e"), scaled = FALSE)
grid.draw(venn)

```

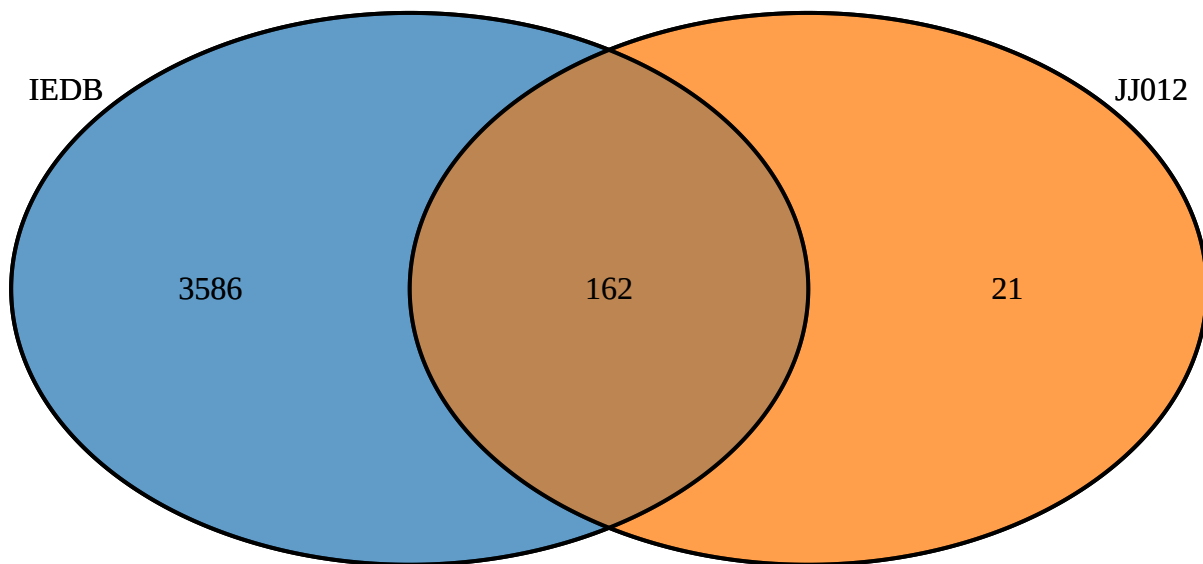


Figure 1: Venn plot JJ012 (MHCI)

CH

```
grid.newpage()
venn <- draw.pairwise.venn(area1 = length(pep_iedb_extended_ch),
  area2 = length(df_final_pep_ch$Seq), cross.area = length(df_final_pep_ch$Seq) -
    length(new_pep_ch_mhc1), category = c("IEDB", "CH2879"),
  fill = c("#2C7BB6", "#ff7f0e"), scaled = FALSE)
grid.draw(venn)
```

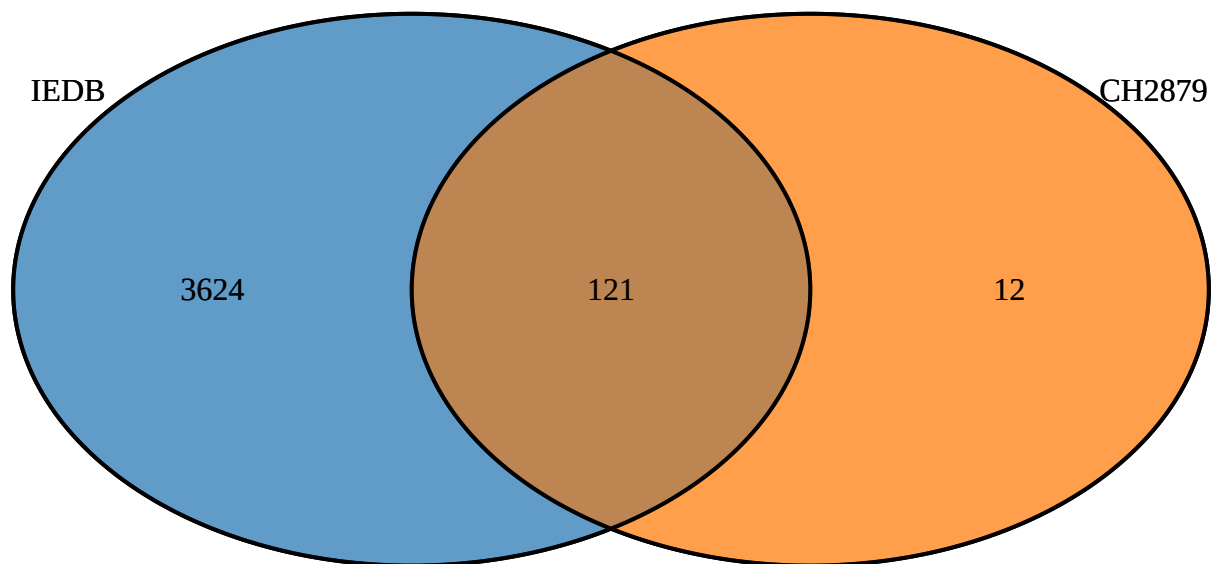


Figure 2: Venn plot CH2879 (MHCI)

II. RNA and protein abundances in cell lines

RNA

```
# Clean data
df_ch2879_tpm <- df_ch2879_tpm[, c("Gene_ID", "Gene_Name", "TPM")]
df_jj012_tpm <- df_jj012_tpm[, c("Gene_ID", "Gene_Name", "TPM")]

# Rename
colnames(df_ch2879_tpm)[3] <- "TPM_CH2879"
colnames(df_jj012_tpm)[3] <- "TPM_JJ012"

# Merge
df_merge <- merge(df_ch2879_tpm, df_jj012_tpm, by = c("Gene_ID",
  "Gene_Name"), all = TRUE)

# Select CTA
df_cta <- df_merge[df_merge$Gene_Name %in% l_CTA, ]
rownames(df_cta) <- df_cta$Gene_Name
df_cta <- df_cta[, c("TPM_CH2879", "TPM_JJ012")]

# Compute matrix in log2 for heatmap
mat_clust_rna <- log2(df_cta + 1)
rna_genes <- rownames(mat_clust_rna)

# Annotation
cluster_colors <- c(A = "#3498DB", B = "#2ECC71", C = "#E74C3C")
colors <- colorRampPalette(c("blue", "white", "red"))(100)
l_clust <- l_clust[match(rownames(mat_clust_rna), l_clust$SYMBOL),
  ]

row_annot_rna <- rowAnnotation(`CTAs cluster` = l_clust$Cluster,
  col = list(`CTAs cluster` = cluster_colors))
```

Protein abundances

```
# Select CTAs
df_prot <- df_prot[df_prot$Gene_name %in% l_CTA, ]

# Prepare data
df_prot_log2 <- df_prot
df_prot_log2[, -1] <- log2(df_prot_log2[, -1] + 1)
mat_prot <- as.matrix(df_prot_log2[, -1])
rownames(mat_prot) <- df_prot_log2$Gene_name
```

Heatmap

```
# Creat df aligned
prot_df_aligned <- data.frame(matrix(NA, nrow = length(rna_genes),
  ncol = ncol(mat_prot)))
rownames(prot_df_aligned) <- rna_genes
colnames(prot_df_aligned) <- colnames(mat_prot)

# Verify common genes
common_genes <- intersect(rna_genes, rownames(mat_prot))
prot_df_aligned[common_genes, ] <- mat_prot[common_genes, ]

# Heatmap for RNA
ht_rna_clust <- Heatmap(as.matrix(mat_clust_rna[, c("TPM_JJ012",
  "TPM_CH2879")]), name = "CTAs expression\n(Log2(TPM + 1))",
  cluster_rows = TRUE, cluster_columns = FALSE, left_annotation = row_annot_rna,
  col = colorRamp2(seq(min(mat_clust_rna), max(mat_clust_rna),
    length.out = 100), colors), column_labels = c("JJ012",
    "CH2879"), row_names_gp = gpar(fontsize = 6))

# Heatmap for prot
ht_prot_clust <- Heatmap(as.matrix(prot_df_aligned[, c("abundances_JJ012_noINF",
  "abundances_CH2879_noINF")]), name = "CTAs protein abundance\n(Log2(Abundance + 1))",
  cluster_rows = TRUE, cluster_columns = FALSE, na_col = "blue",
  col = colorRamp2(seq(min(prot_df_aligned, na.rm = TRUE),
    max(prot_df_aligned, na.rm = TRUE), length.out = 100),
    colors), column_labels = c("JJ012", "CH2879"), row_names_gp = gpar(fontsize = 6))

# svg('../results/combined_heatmap_rna_prot_clust_v2.svg',
# height = 10, width = 10)
ht_rna_clust + ht_prot_clust

# dev.off()
```


III. Peptide ranks and scatter plots

```
# Select dominant HLA type
df_aff_ch_hlaa0101 <- df_aff_ch[df_aff_ch$Allele == "HLA-A0101" &
  df_aff_ch$Peptide %in% df_ch_mhc1$Seq, c("Peptide", "Affinity",
  "Rank")]

df_aff_jj_hlaa0201 <- df_aff_jj[df_aff_jj$Allele == "HLA-A0201" &
  df_aff_jj$Peptide %in% df_jj_mhc1$Seq, c("Peptide", "Affinity",
  "Rank")]
```

CH2879

Scatter plot from Fig1E

```
# Add known pep
pep_commons <- intersect(df_final_pep_ch$Seq, pep_iedb_mhc1_cta)

# Reunite known peptides
pep_commons2 <- names(grep_ch)[grep_ch]
pep_known_all <- union(pep_commons, pep_commons2)

# Dataframe with know CTA peptides and new peptides
df_iedb_new_pep_ch <- df_ch_mhc1 %>%
  filter(Seq %in% new_pep_ch_mhc1 | Seq %in% pep_known_all) %>%
  distinct(Seq, .keep_all = TRUE)

# Prepare df for scatter
df_iedb_new_pep_ch_anno <- prep_df(df_iedb_new_pep_ch)
df_merge <- merge(df_iedb_new_pep_ch, df_iedb_new_pep_ch_anno,
  by.x = "Prot_accessions", by.y = "Proteins")

# Add rank information
df_pep_ch_hlaa0101 <- df_merge[df_merge$Seq %in% df_aff_ch_hlaa0101$Peptide,
]
df_pep_ch_hlaa0101 <- merge(df_pep_ch_hlaa0101, df_aff_ch_hlaa0101,
  by.x = "Seq", by.y = "Peptide")
df_scatter_ch <- merge(df_scatter_hr, df_pep_ch_hlaa0101, by.x = "SYMBOL",
  by.y = "Genes")

# Plot
df_scatter_ch <- df_scatter_ch %>%
  mutate(Status = case_when(Seq %in% new_pep_ch_mhc1 ~ "New",
    Seq %in% pep_known_all ~ "Known"))

# Agregate per gene, if one is new => orange
df_scatter_ch_gene <- df_scatter_ch %>%
  group_by(SYMBOL, Mean_expression_tissues, Intensity, HR) %>%
  summarise(Status = ifelse(any(Status == "New"), "New", "Known"),
    Rank = min(Rank), .groups = "drop")
```

```

# Labeling genes
df_labels <- df_scatter_ch_gene %>%
  filter(Mean_expression_tissues < quantile(df_scatter_hr$Mean_expression_tissues,
    0.75), Intensity > median(df_scatter_hr$Intensity)) %>%
  mutate(is_sb = Rank <= 0.5)

# For legends
df_legend <- data.frame(Mean_expression_tissues = c(NA, NA),
  Intensity = c(NA, NA), Status = c("Strong binders", "Others"))

# Plot
p <- ggplot(df_scatter_ch_gene, aes(x = Mean_expression_tissues,
  y = Intensity, color = Status, text = paste0("Gene: ", SYMBOL,
    "<br>Status: ", Status, "<br>Mean expression: ", round(Mean_expression_tissues,
    3), "<br>Intensity: ", round(Intensity, 3), "<br>HR: ",
    round(HR, 3), "<br>Rank: ", round(Rank, 3)))) + geom_point(size = 2,
  alpha = 0.8) + geom_point(data = df_legend, aes(x = Mean_expression_tissues,
  y = Intensity, color = Status), size = 0, alpha = 0, inherit.aes = FALSE) +
  scale_color_manual(values = c(Known = "#1f77b4", New = "#ff7f0e",
    `Strong binders` = "black", Others = "black"), labels = c(Known = "Known peptides",
    New = "Novel peptides", `Strong binders` = expression(bold("Strong binders (bold)")),
    Others = "Others"), breaks = c("Known", "New", "Strong binders",
    "Others")) + guides(color = guide_legend(override.aes = list(size = c(2,
  2, 0, 0), alpha = c(1, 1, 0, 0)))) + scale_x_log10(breaks = c(0.001,
  0.01, 0.1, 1, 10), labels = c("0.001", "0.01", "0.1", "1",
  "10")) + geom_hline(yintercept = median(df_scatter_hr$Intensity),
  linetype = "dashed", color = "black") + annotate("text",
  x = max(df_scatter_ch_gene$Mean_expression_tissues), y = median(df_scatter_hr$Intensity) -
    0.2, label = "Median\nexpression", size = 2.5) + geom_vline(xintercept = quantile(df_scatter_hr$
  0.75), linetype = "dashed", color = "black") + annotate("text",
  x = quantile(df_scatter_hr$Mean_expression_tissues, 0.75) +
    0.8, y = max(df_scatter_ch_gene$Intensity) + 0.2, label = "75th percentile",
  size = 2.5) + geom_text_repel(data = df_labels, aes(label = SYMBOL,
  fontface = ifelse(is_sb, "bold", "plain"), size = 3), color = "black",
  max.overlaps = 20, box.padding = 0.4, segment.color = "black",
  segment.size = 0.3, min.segment.length = 0) + scale_size_identity() +
  theme_minimal() + labs(x = "CTAs expression in normal tissues (Log10 TPM mean)",
  y = "CTAs expression in CHS (mean intensities)", color = "CH2879 HLA-A*01:01\npeptides")

```

p

```

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).

```

```

# ggplotly(p, tooltip = 'text')
# ggsave('../results/scatter_plot_ch.svg', plot = p, width
# = 20, height = 15, units = 'cm')

```

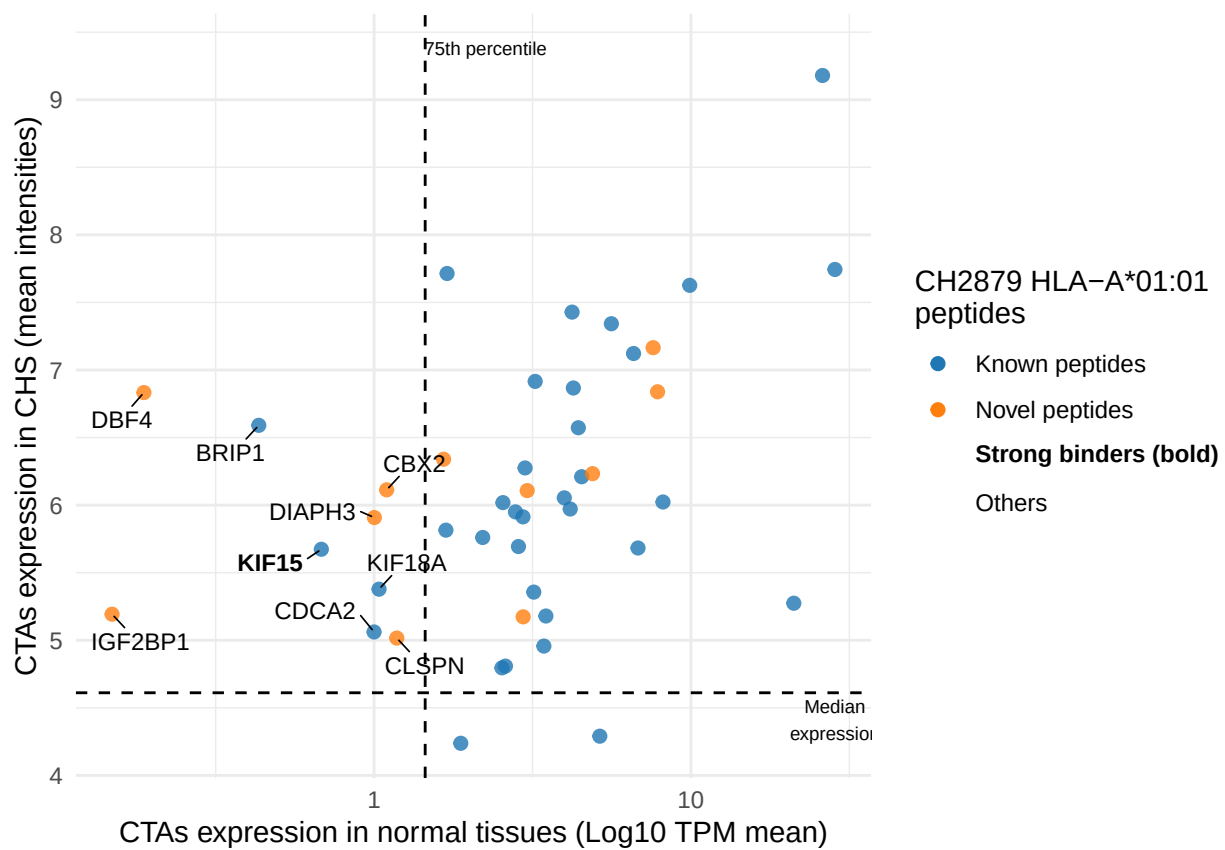


Figure 4: Scatter plot from Fig1E (CH2879)

Rank scatter plot

```
# Count number of peptides per rank
df_count <- df_scatter_ch %>%
  ungroup() %>%
  mutate(Status = as.character(Status)) %>%
  group_by(Rank) %>%
  summarise(nb_pep = n(), is_new = any(Status == "New"))

# Scatter plot
p <- ggplot(df_count, aes(x = nb_pep, y = Rank, color = is_new)) +
  geom_point(size = 2) + scale_color_manual(values = c(`FALSE` = "grey30",
`TRUE` = "#ff7f0e"), labels = c(`FALSE` = "Known peptides",
`TRUE` = "Novel peptides")) + geom_hline(yintercept = c(0.5,
2), linetype = "dashed", color = "black") + annotate("text",
x = max(df_count$nb_pep), y = 0.5 * 0.8, label = "Strong binders",
hjust = 1, color = "black") + annotate("text", x = max(df_count$nb_pep),
y = 2 * 0.8, label = "Weak binders", hjust = 1, color = "black") +
labs(x = "CH2879 HLA-A*01:01 peptides number", y = "%Rank",
color = "Category") + scale_y_log10(breaks = c(0.01,
0.1, 1, 10, 100), labels = c("0.01", "0.1", "1", "10", "100")) +
scale_x_continuous(breaks = seq(0, max(df_count$nb_pep),
by = 2)) + theme_minimal()
```

p

```
# ggsave('../results/scatter_ch_a01.svg', plot = p)
```

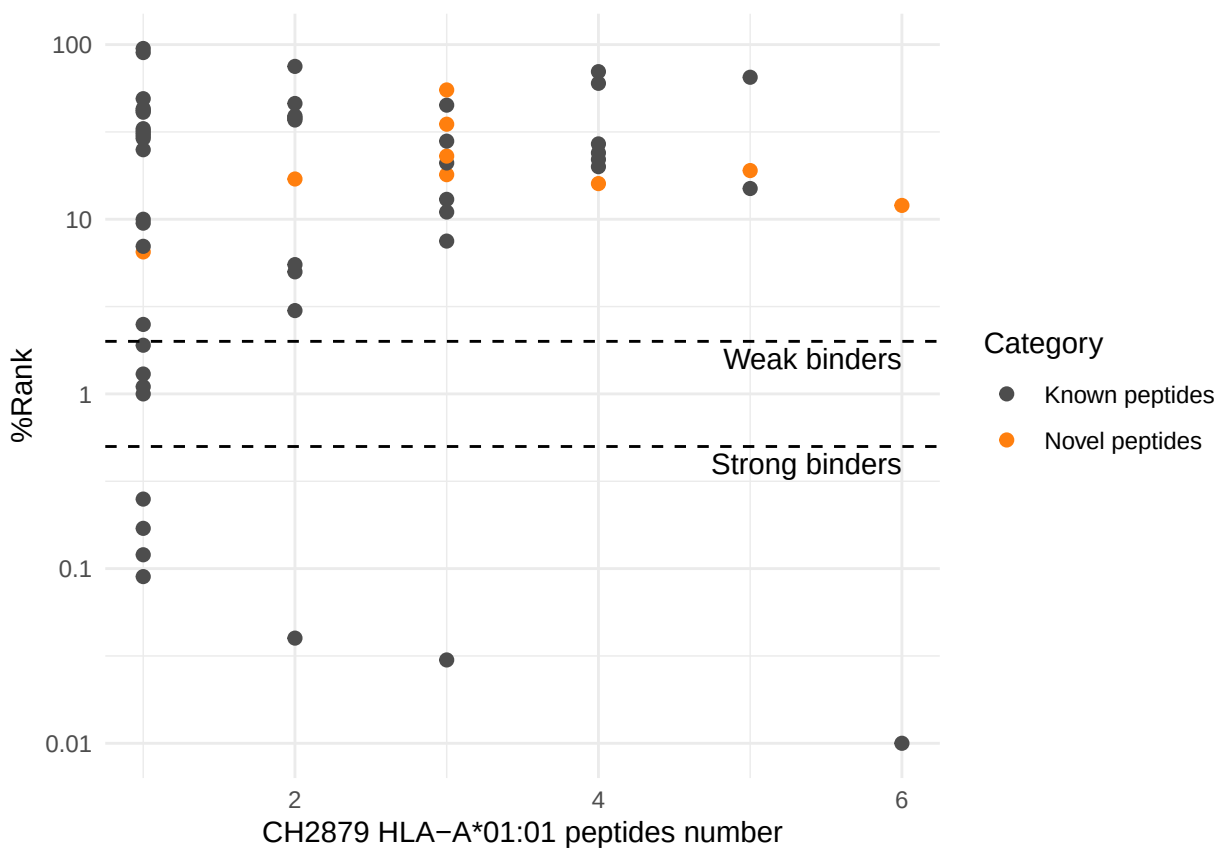


Figure 5: Rank scatter plot (CH2879)

JJ012

Scatter plot from Fig1E

```
# Add known pep
pep_commons <- intersect(df_final_pep_jj$Seq, pep_iedb_mhc1_cta)

# Reunite known peptides
pep_commons2 <- names(grep_jj)[grep_jj]
pep_known_all <- union(pep_commons, pep_commons2)

# Dataframe with know CTA peptides and new peptides
df_iedb_new_pep_jj <- df_jj_mhc1 %>%
  filter(Seq %in% new_pep_jj_mhc1 | Seq %in% pep_known_all) %>%
  distinct(Seq, .keep_all = TRUE)

# Prepare df for scatter
df_iedb_new_pep_jj_anno <- prep_df(df_iedb_new_pep_jj)
df_merge <- merge(df_iedb_new_pep_jj, df_iedb_new_pep_jj_anno,
  by.x = "Prot_accessions", by.y = "Proteins")

# Add rank information
df_pep_jj_hlaa0201 <- df_merge[df_merge$Seq %in% df_aff_jj_hlaa0201$Peptide,
]
df_pep_jj_hlaa0201 <- merge(df_pep_jj_hlaa0201, df_aff_jj_hlaa0201,
  by.x = "Seq", by.y = "Peptide")

# Prepare data for scatter
df_scatter_jj <- merge(df_scatter_hr, df_pep_jj_hlaa0201, by.x = "SYMBOL",
  by.y = "Genes")

# Plot
df_scatter_jj <- df_scatter_jj %>%
  mutate(Status = case_when(Seq %in% new_pep_jj_mhc1 ~ "New",
    Seq %in% pep_known_all ~ "Known"))

# Agregate per gene, if one is new => orange
df_scatter_jj_gene <- df_scatter_jj %>%
  group_by(SYMBOL, Mean_expression_tissues, Intensity, HR) %>%
  summarise(Status = ifelse(any(Status == "New"), "New", "Known"),
    Rank = min(Rank), .groups = "drop")

# Labeling genes
df_labels <- df_scatter_jj_gene %>%
  filter(Mean_expression_tissues < quantile(df_scatter_hr$Mean_expression_tissues,
    0.75), Intensity > median(df_scatter_hr$Intensity)) %>%
  mutate(is_sb = Rank <= 0.5)

# Plot
p <- ggplot(df_scatter_jj_gene, aes(x = Mean_expression_tissues,
  y = Intensity, color = Status, text = paste0("Gene: ", SYMBOL,
    "<br>Status: ", Status, "<br>Mean expression: ", round(Mean_expression_tissues,
```

```

    3), "<br>Intensity: ", round(Intensity, 3), "<br>HR: ",
    round(HR, 3), "<br>Rank: ", round(Rank, 3))) + geom_point(size = 2,
alpha = 0.8) + geom_point(data = df_legend, aes(x = Mean_expression_tissues,
y = Intensity, color = Status), size = 0, alpha = 0, inherit.aes = FALSE) +
scale_color_manual(values = c(Known = "#1f77b4", New = "#ff7f0e",
`Strong binders` = "black", Others = "black"), labels = c(Known = "Known peptides",
New = "Novel peptides", `Strong binders` = expression(bold("Strong binders (bold)")),
Others = "Others"), breaks = c("Known", "New", "Strong binders",
"Others")) + guides(color = guide_legend(override.aes = list(size = c(2,
2, 0, 0), alpha = c(1, 1, 0, 0)))) + geom_hline(yintercept = median(df_scatter_hr$Intensity),
linetype = "dashed", color = "black") + annotate("text",
x = max(df_scatter_jj_gene$Mean_expression_tissues), y = median(df_scatter_hr$Intensity) -
0.2, label = "Median\nexpression", size = 2.5) + geom_vline(xintercept = quantile(df_scatter_hr$
0.75), linetype = "dashed", color = "black") + annotate("text",
x = quantile(df_scatter_hr$Mean_expression_tissues, 0.75) +
0.8, y = max(df_scatter_jj_gene$Intensity) + 0.2, label = "75th percentile",
size = 2.5) + geom_text_repel(data = df_labels, aes(label = SYMBOL,
fontface = ifelse(is_sb, "bold", "plain"), size = 3), color = "black",
max.overlaps = 20, box.padding = 0.4, segment.color = "black",
segment.size = 0.3, min.segment.length = 0) + scale_size_identity() +
scale_x_log10(breaks = c(0.001, 0.01, 0.1, 1, 10), labels = c("0.001",
"0.01", "0.1", "1", "10")) + theme_minimal() + labs(x = "CTAs expression in normal tissues (Log
y = "CTAs expression in CHS (mean intensities)", color = "JJ012 HLA-A*02:01\npeptides")

```

p

```

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).

```

```

# ggsave('../results/scatter_plot_jj.svg', plot = p, width
# = 20, height = 15, units = 'cm')

```

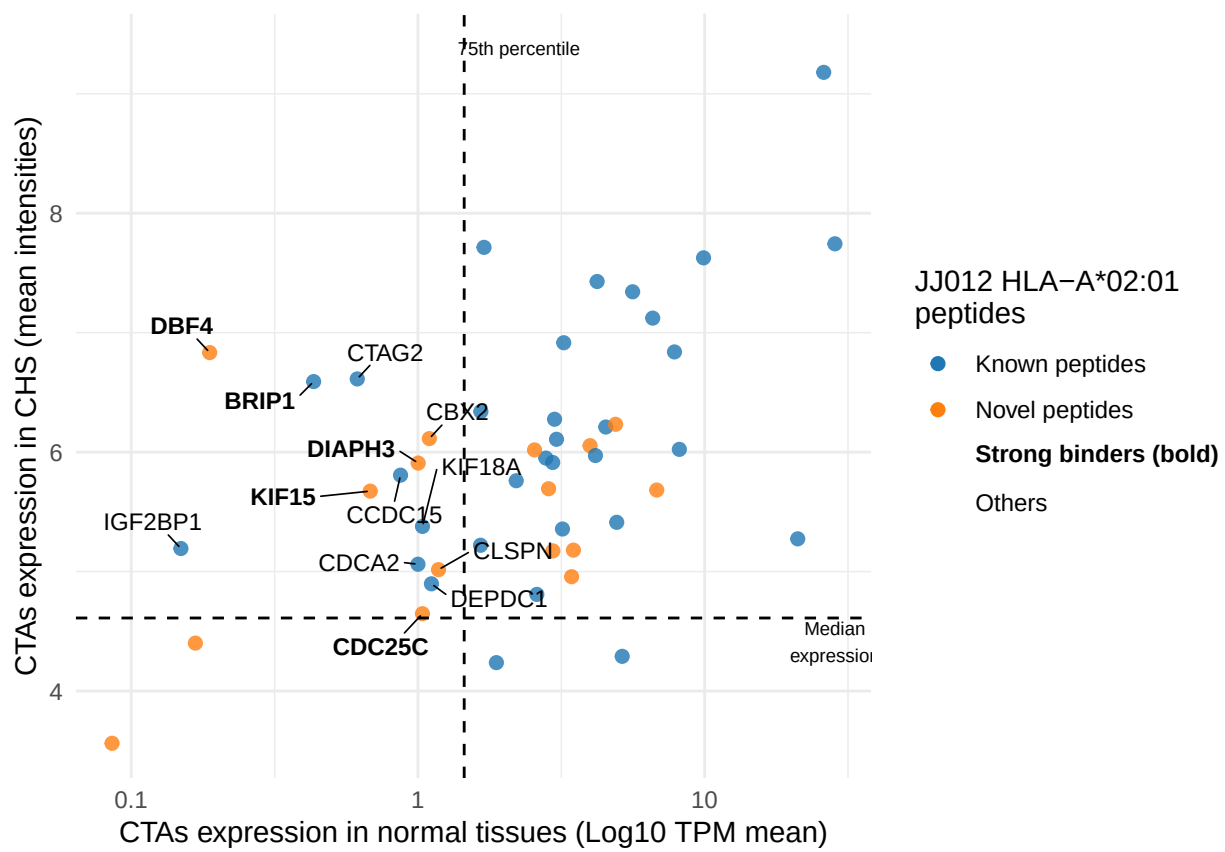


Figure 6: Scatter plot from Fig1E (JJ012)

Rank scatter plot

```
# Count number of peptides per rank
df_count <- df_scatter_jj %>%
  ungroup() %>%
  mutate(Status = as.character(Status)) %>%
  group_by(Rank) %>%
  summarise(nb_pep = n(), is_new = any(Status == "New"))

# Scatter plot
p <- ggplot(df_count, aes(x = nb_pep, y = Rank, color = is_new)) +
  geom_point(size = 2) + scale_color_manual(values = c(`FALSE` = "grey30",
  `TRUE` = "#ff7f0e"), labels = c(`FALSE` = "Known peptides",
  `TRUE` = "Novel peptides")) + geom_hline(yintercept = c(0.5,
  2), linetype = "dashed", color = "black") + annotate("text",
  x = max(df_count$nb_pep), y = 0.5 * 0.8, label = "Strong binders",
  hjust = 1, color = "black") + annotate("text", x = max(df_count$nb_pep),
  y = 2 * 0.8, label = "Weak binders", hjust = 1, color = "black") +
  labs(x = "JJ012 HLA-A*02:01 peptides number", y = "%Rank",
  color = "Category") + scale_y_log10(breaks = c(0.01,
  0.1, 1, 10, 100), labels = c("0.01", "0.1", "1", "10", "100")) +
  theme_minimal()
p
```

```
# ggsave('../results/scatter_jj_a02.svg', plot = p)
```

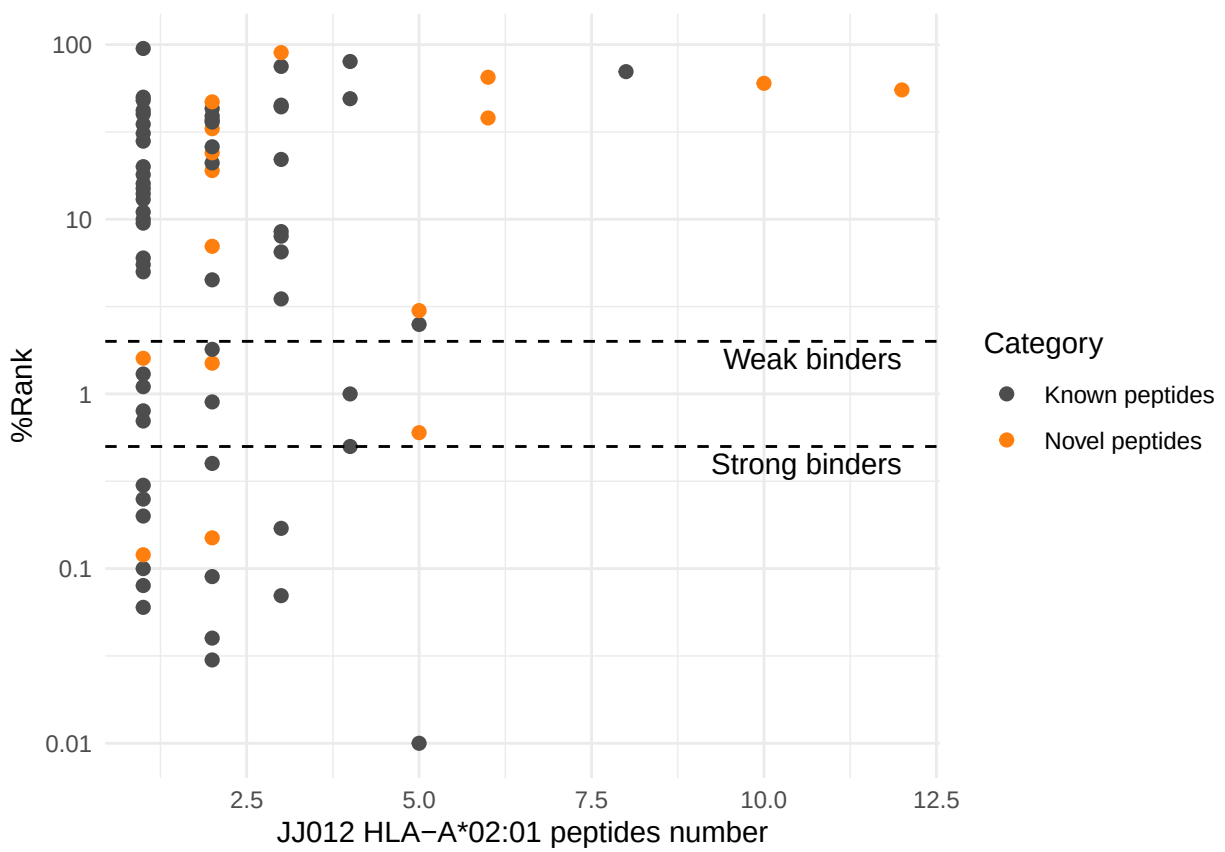


Figure 7: Rank scatter plot (JJ012)

IV. Supplementary data 2

```
# Select data
df_ch_supp <- df_scatter_ch[df_scatter_ch$Intensity > median(df_scatter_hr$Intensity) &
  df_scatter_ch$Mean_expression_tissues < quantile(df_scatter_hr$Mean_expression_tissues,
    0.75), c("SYMBOL", "Intensity", "Mean_expression_tissues",
    "Cluster", "Seq", "Affinity", "Rank", "Status")]
df_jj_supp <- df_scatter_jj[df_scatter_jj$Intensity > median(df_scatter_hr$Intensity) &
  df_scatter_jj$Mean_expression_tissues < quantile(df_scatter_hr$Mean_expression_tissues,
    0.75), c("SYMBOL", "Intensity", "Mean_expression_tissues",
    "Cluster", "Seq", "Affinity", "Rank", "Status")]

# Write in excel file
write_xlsx(list(JJ012_peptides_cadran = df_jj_supp, CH2879_peptides_cadran = df_ch_supp),
  path = "../results/fig4_supp.xlsx")
```